AI600 Assignment 3 Report

Course Context

Lahore University of Management Sciences, School of Science and Engineering, Department of Electrical Engineering. This report covers the analytical Toy-CNN questions and the programming experiments for standard MNIST, Colored-MNIST, and STL-10. Target GitHub repository: https://github.com/abdull2325/assignment3-dl.

Execution Status

Standard MNIST, Colored-MNIST, and STL-10 were all executed successfully. For STL-10, the required train/test binaries were extracted from the official archive before the large unlabeled split completed downloading, which was sufficient for Task 2.

Key Metrics

MNIST test accuracy: 98.76%. Colored-MNIST biased test accuracy: 98.49%. Colored-MNIST unbiased test accuracy: 75.52%. STL-10 test accuracy: 79.61%. Custom CNN parameter count: 42722.

Analytical Solutions

Q1 Forward Pass

The convolution pre-activation map is $Z = [[2, 2], [-1, 3]]$. After ReLU, $A = [[2, 2], [0, 3]]$. Flattening gives $[2, 2, 0, 3]^T$. The scalar prediction is $Y = 2.500$ and the loss is $L = 0.125$.

Q1 Backward Pass

The loss gradient with respect to the fully connected weights is $dL/dV = [-1, -1, 0, -1.5]$. The convolution-filter gradient is $dL/dW = [[0.25, -1.25], [0.5, -0.5]]$. The largest-magnitude entry is the top-right weight, so gradient descent increases that weight the most.

Q3 Pooling and Normalization

With global max pooling, $Y = \max(A) = 3$, which matches the target, so the loss is zero and all input gradients are zero in this numeric example. For layer normalization on A, the mean is 1.75, the variance is 1.1875, and the normalized map is approximately $[[0.2294, 0.2294], [-1.6059, 1.1471]]$.

Practice Questions

For completeness, the input gradient in Q2 is $dL/dX = [[-0.5, 0.5, 0], [0.5, -1.25, 0.5], [0, 0.25, -0.25]]$, so an adversary would decrease the center pixel to increase the loss. In the residual variant, $A_{res} = [[3, 4], [0, 4]]$, $Y_{res} = 2.0$, and dL/dX_{11} becomes -2.0, illustrating the gradient-highway effect of skip connections.

Programming Analysis

Task 1A: Standard MNIST

The custom CNN uses three convolutional layers and two linear layers while staying under the 50,000-parameter budget with 42722 trainable parameters. Final test accuracy is 98.76%. The model generalizes well: training accuracy (0.982) and validation accuracy (0.986) stay close, with an absolute gap of 0.003.

Task 1B: Colored-MNIST

The same architecture was adapted to RGB input. It reached 98.49% on the biased test set but only 75.52% on the unbiased test set, confirming shortcut learning through color. The network attains 0.985 accuracy on the biased test split but falls to 0.755 on the unbiased split, a drop of 0.230. Cross-entropy minimization rewards whichever feature most reliably reduces training loss. Because color is highly correlated with the label in the biased training data and is easier to separate than shape, the optimizer converges to color-sensitive filters first.

Strategies to Reduce Shortcut Learning

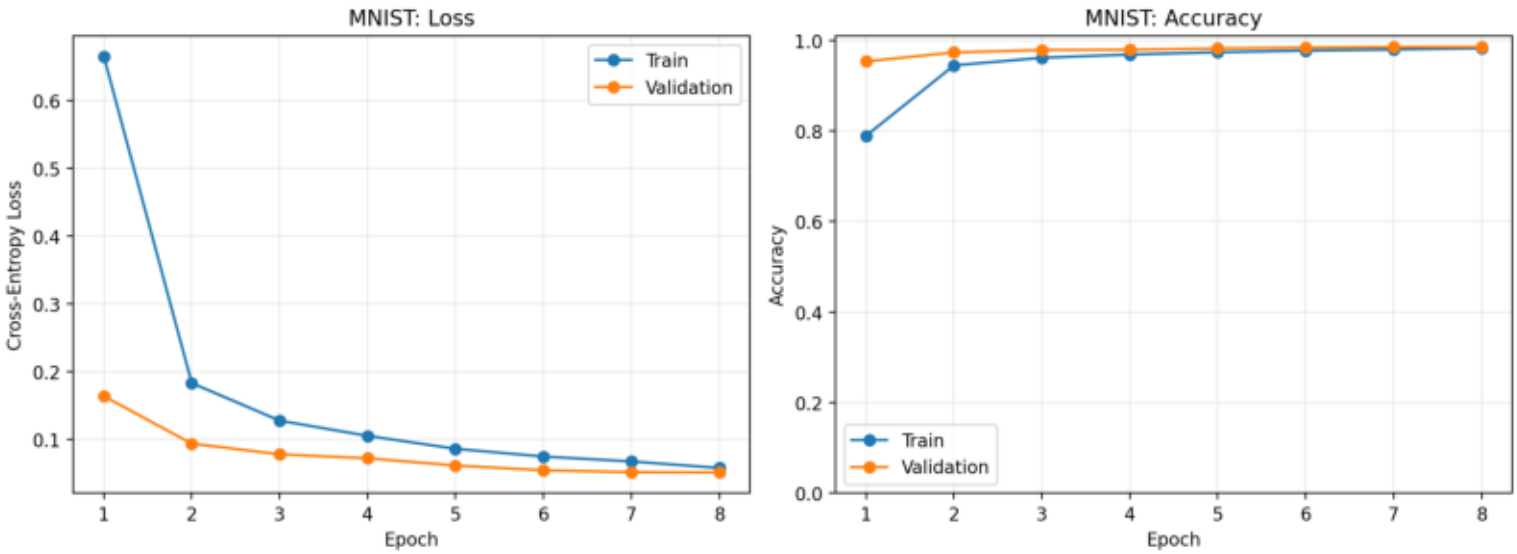
Useful interventions include grayscale conversion, heavy color jitter, randomized recoloring, balanced color-label pairings, and invariant-learning objectives such as Group DRO or IRM.

Task 2: Transfer Learning and GradCAM

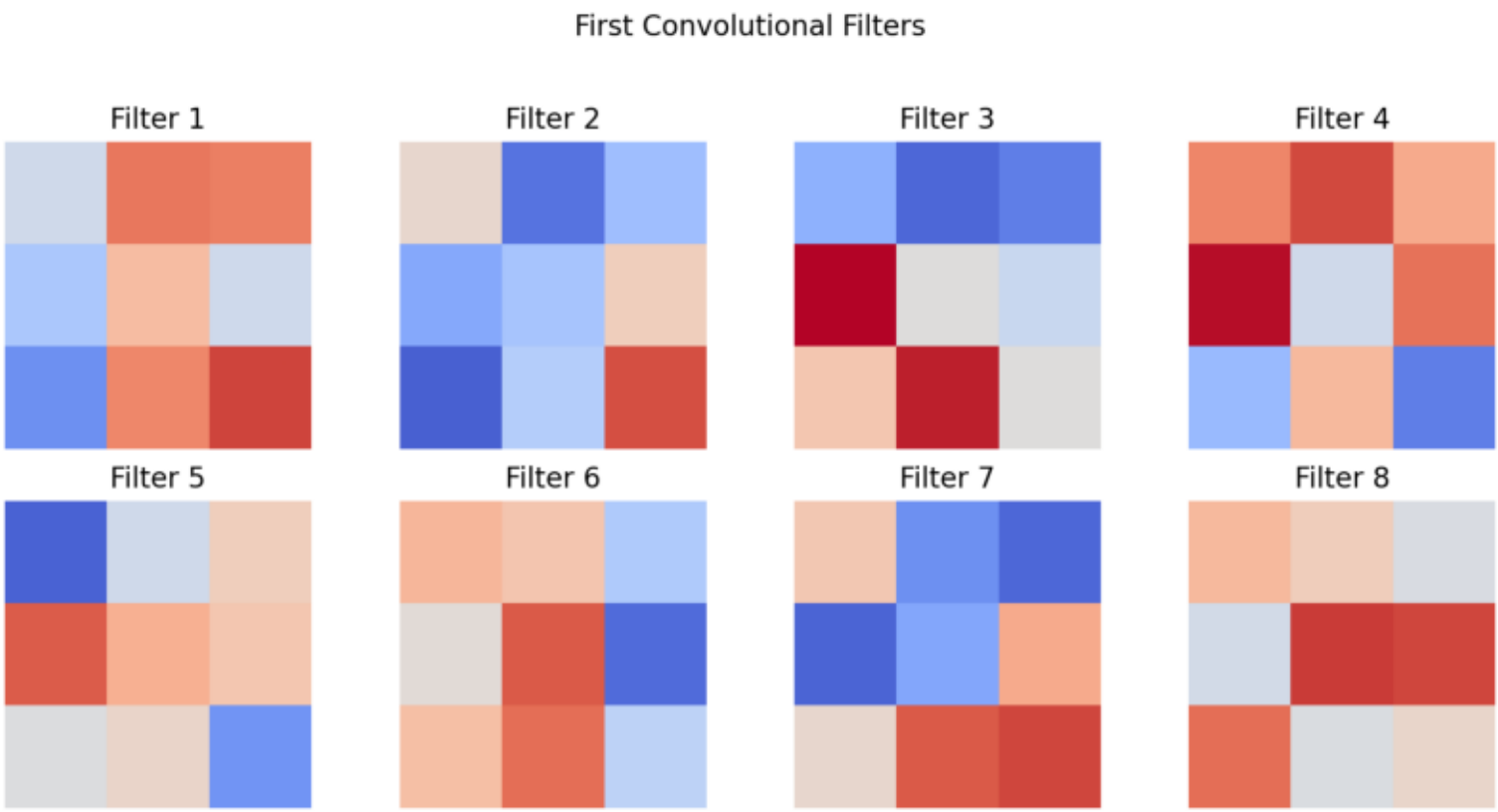
A frozen-backbone pretrained ResNet-18 was fine-tuned on STL-10 by training only the classification head. The final STL-10 test accuracy is 79.61%. The GradCAM overlays show that correct predictions usually center on the main object, while mistakes often over-emphasize a localized part such as a face, branch crossing, or ambiguous body contour.

MNIST Results

Training and validation curves

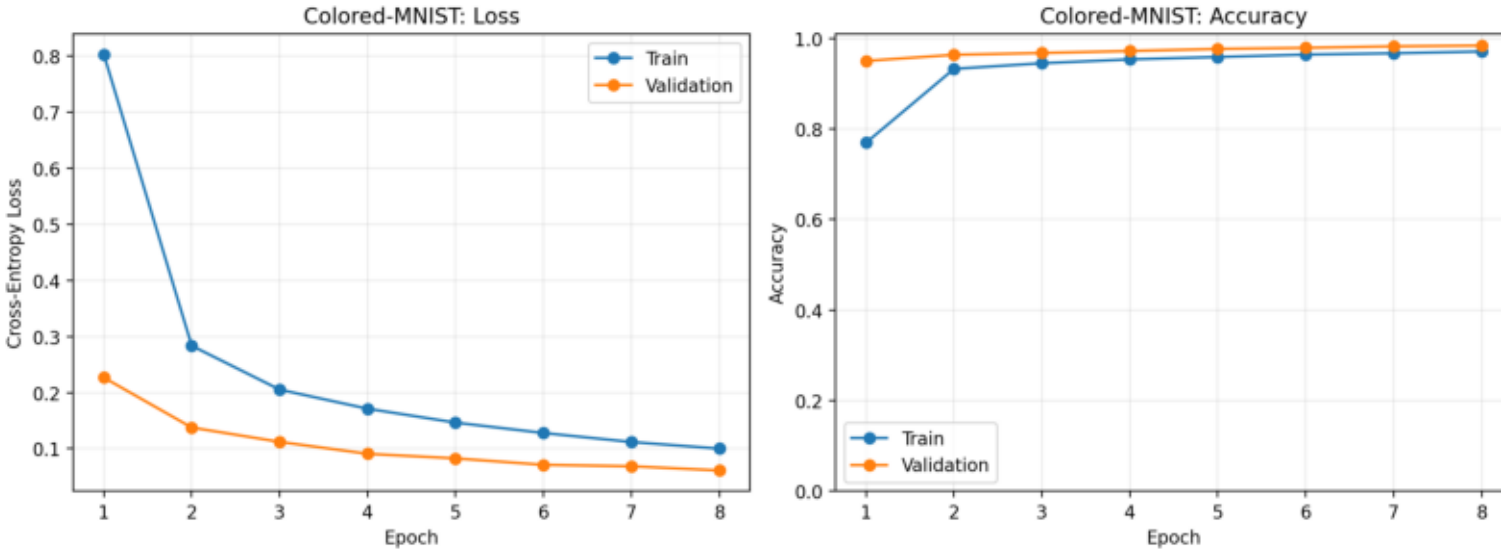


First-layer convolutional filters

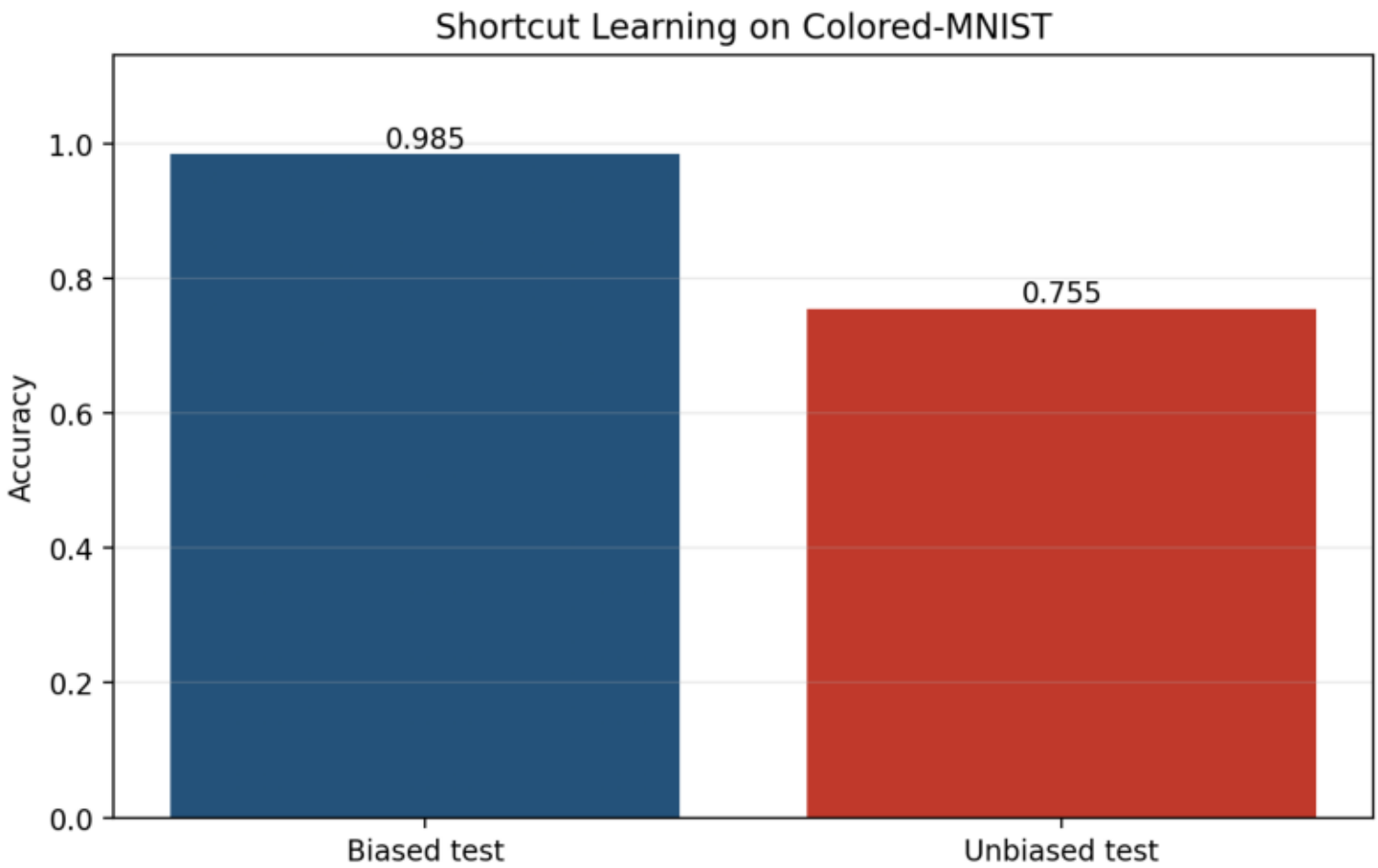


Colored-MNIST Results

Training and validation curves

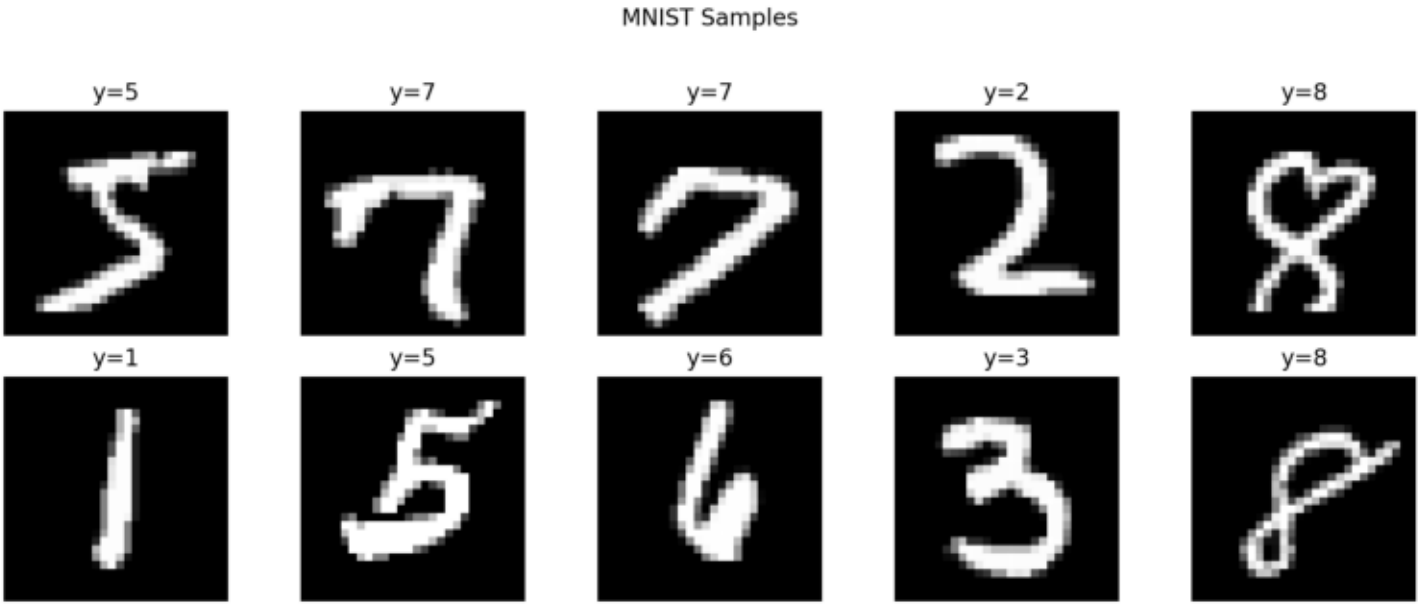


Accuracy gap between biased and unbiased tests

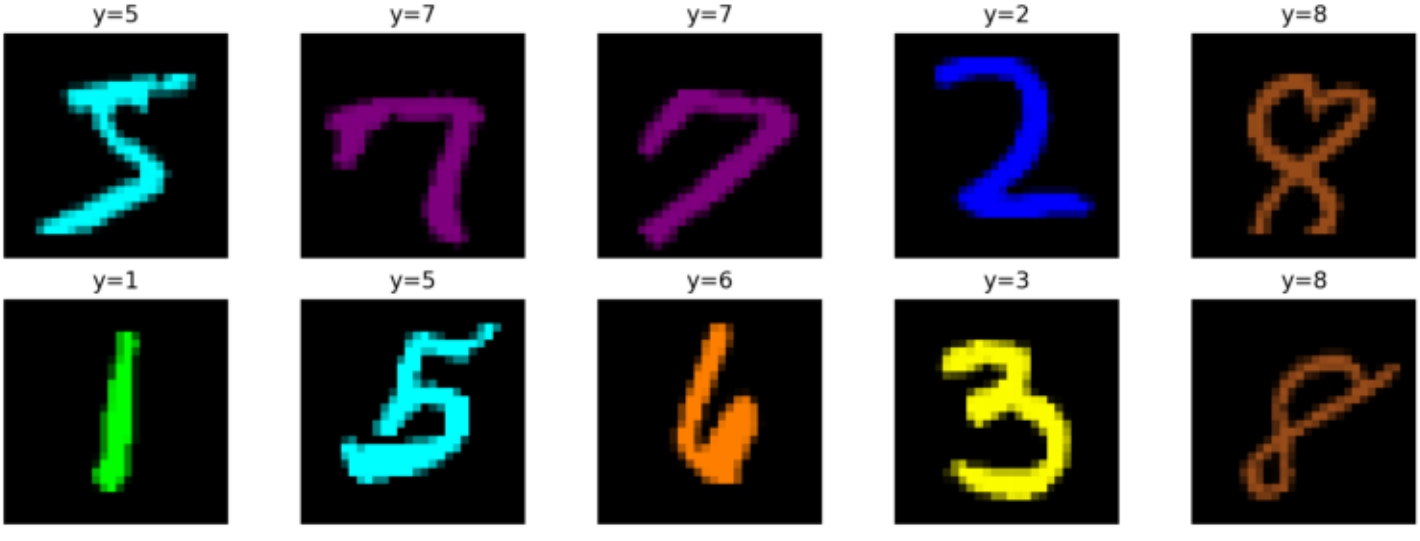


Sample Visualizations

MNIST sample grid



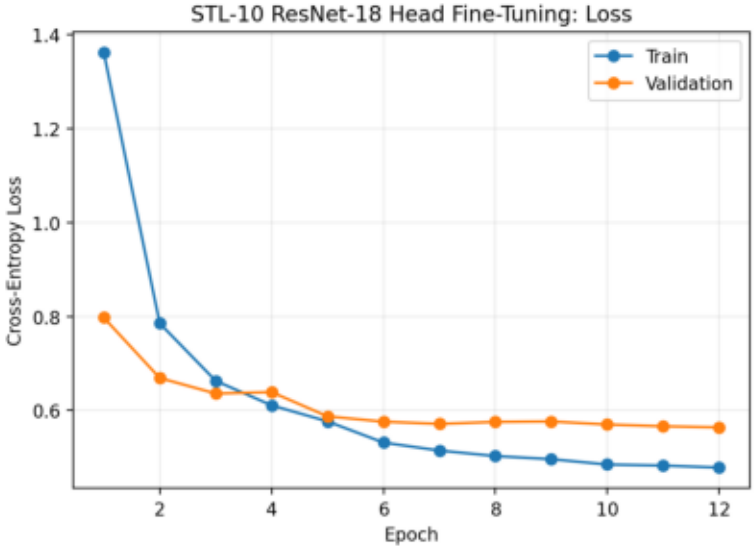
Colored-MNIST biased training samples

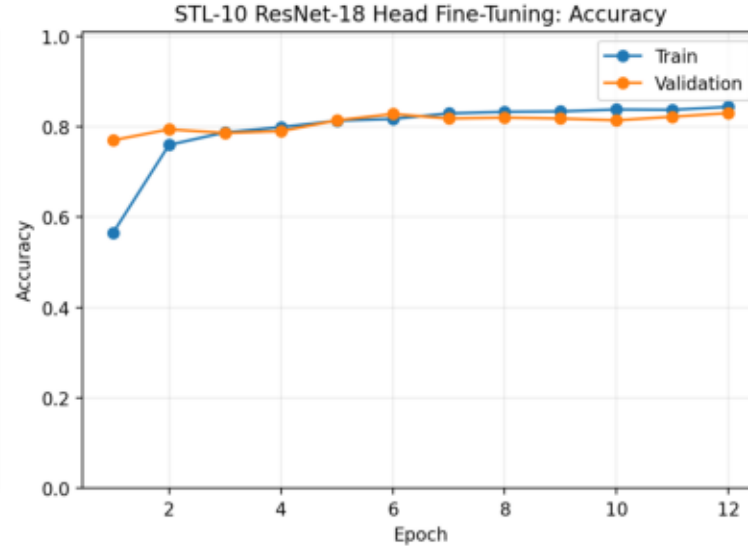


Colored-MNIST unbiased test samples

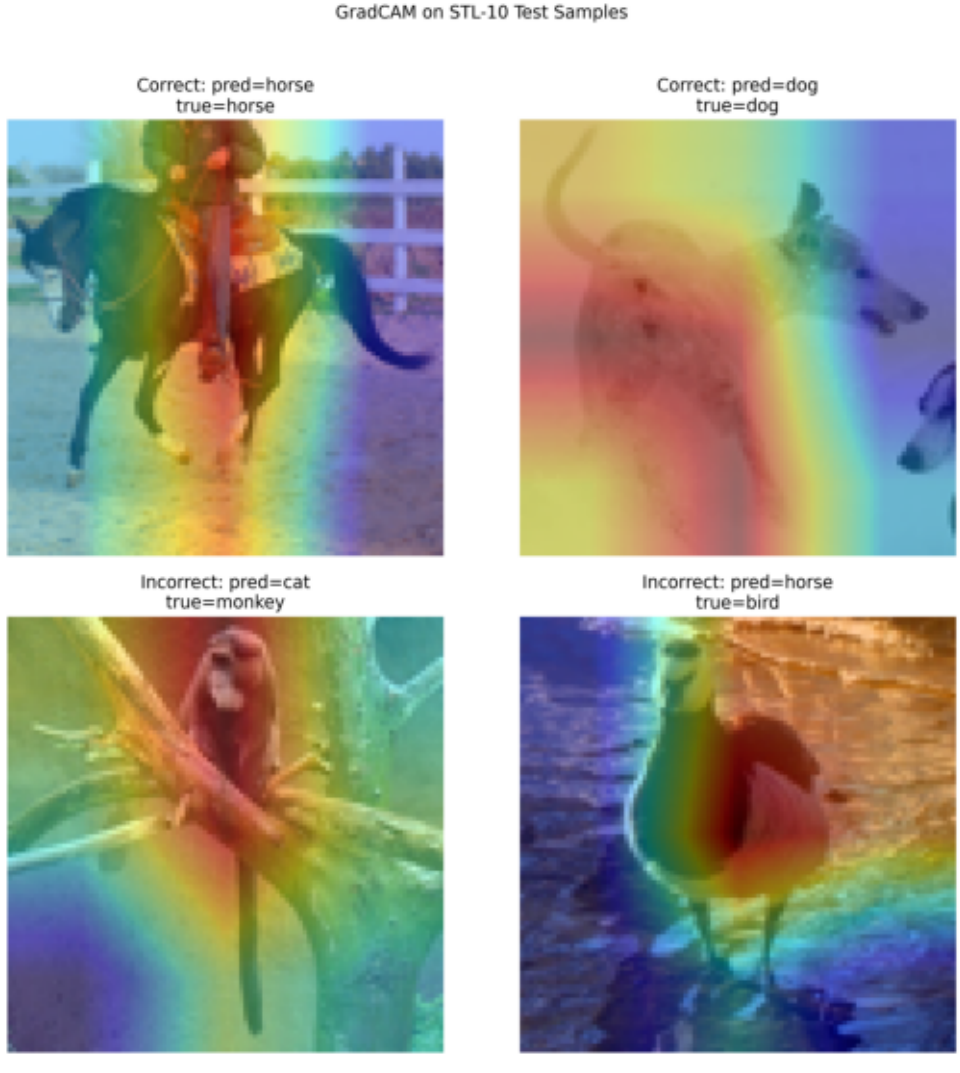
STL-10 Results

Frozen-backbone ResNet-18 training curves





GradCAM overlays on correct and incorrect test predictions



AI Policy Disclosure

Prompt Used

you have to complete the following assessmetn make sure things are aligned and we are good to go accordingly make sure you complete the assignment along with the visualisation of everything and make sure its completed thoroughly and greatly

Generated Output

The AI-assisted workflow generated analytical derivations, PyTorch training code, plotting utilities, report scaffolding, and written interpretations of the observed results.

Edits Applied

The generated content was edited to match the exact numerical outputs, enforce the parameter budget, incorporate measured metrics from artifacts/results.json, patch the STL-10 loader to use the extracted train/test binaries from the official archive, and add the actual GradCAM-based interpretation from the generated heatmaps.